UAV Insulator Classification

Automated Deep Learning Ground Station using YOLOv8n

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B.Tech Electrical & Electronics Engineering, 2nd Year

Project Overview

Revolutionizing utility inspection through computer vision and autonomous monitoring.

The Manual vs. UAV Paradigm

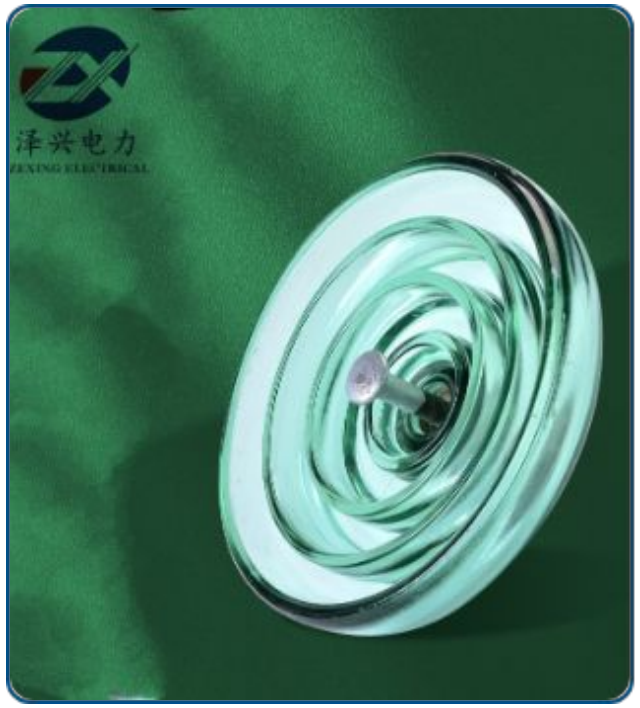
Traditional Challenges

Manual tower climbing and ground-based inspections are hazardous, time-consuming, and prone to human error, with limited reach in rugged terrains.

The AI Solution

UAVs equipped with high-resolution sensors capture critical data, while YOLOv8n models classify materials in near real-time, ensuring grid reliability.

Target Material Classes



Porcelain

Heavy and traditional ceramic materials requiring specific maintenance cycles.

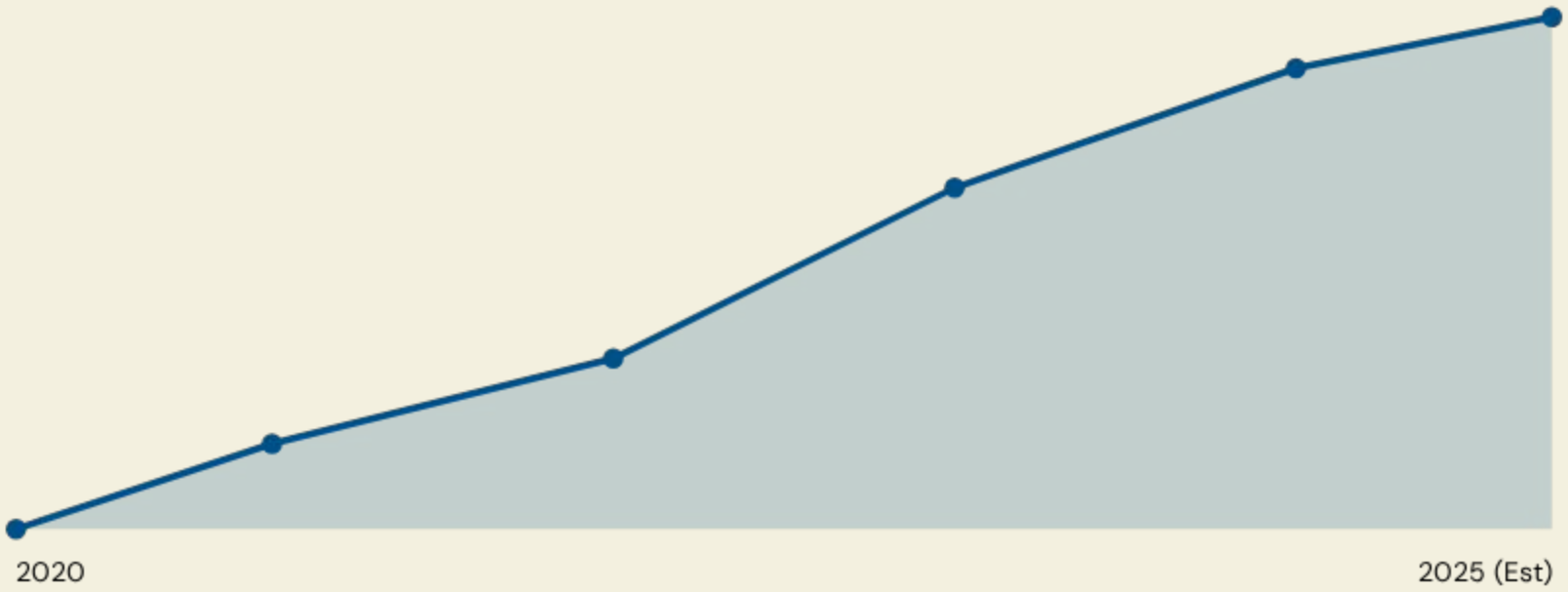
Glass

High dielectric strength and visible self-cleaning properties for polluted areas.

Composite

Lightweight polymer-based sheds with superior pollution performance.

Global Adoption of Drone Inspection



Exponential growth in UAV adoption for utility maintenance driven by safety and operational efficiency.

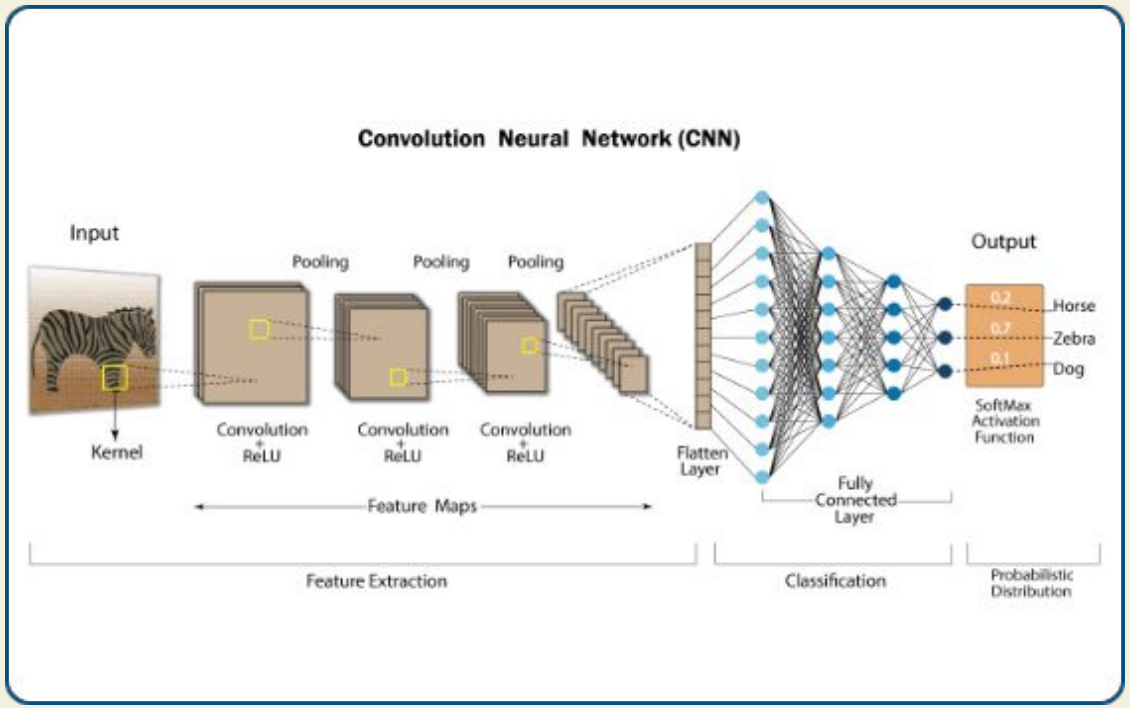
YOLOv8n Technical Architecture

Backbone: Employs **C2f modules** (CSP Bottleneck with 2 convolutions) for efficient cross-stage feature extraction.

Neck: Utilizes **PANet** for enhanced multi-scale feature fusion across top-down and bottom-up paths.

Head: A **Decoupled Head** architecture separating classification and localization tasks.

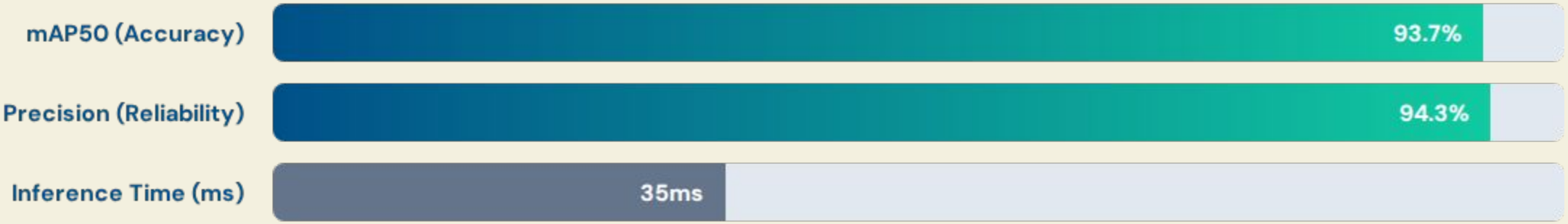
Anchor-Free: Simplifies training and allows for better generalization to various insulator shapes.



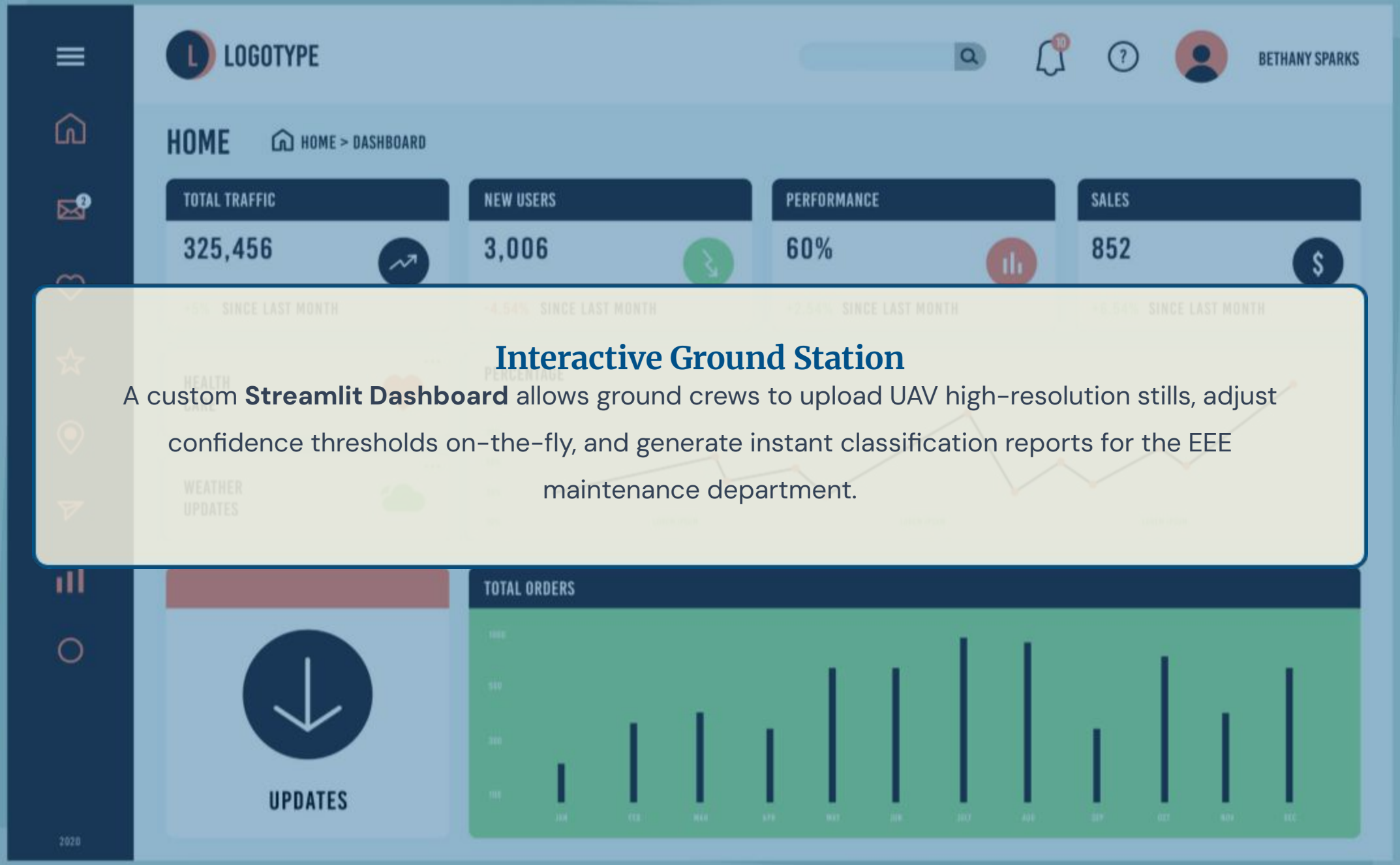
Training Configuration & Process

Parameter	Configuration Value	Rationale
Base Model	YOLOv8n (Nano)	Optimized for i5 CPU inference
Image Size	640px	Balanced detail vs speed
Optimizer	AdamW	Robust weight decay handling
Batch Size	16	Memory management for local setup
Augmentation	Mosaic, Flip, HSV	Improves "Composite" insulator recall

Model Performance Evaluation



Our fine-tuned model demonstrates professional-grade precision, effectively distinguishing materials in complex background noise.



Interactive Ground Station

A custom **Streamlit Dashboard** allows ground crews to upload UAV high-resolution stills, adjust confidence thresholds on-the-fly, and generate instant classification reports for the EEE maintenance department.

The Engineering Tech Stack

Python Core

Utilizing venv for environment isolation and Python 3.13 stability.

Ultralytics

The core YOLOv8 framework for model training and deployment.

Git Workflow

Standardized repository structure for reproducibility and collaboration.

Future Enhancements Roadmap

Defect Detection

Identify cracks, burns, and pollution flashover marks.

GPS Mapping

Extract EXIF data to plot insulators on interactive GIS maps.

Edge TPU

Optimize for NVIDIA Jetson or Coral TPU onboard the UAV.

Full Autonomy

Autonomous flight triggering based on AI detections.